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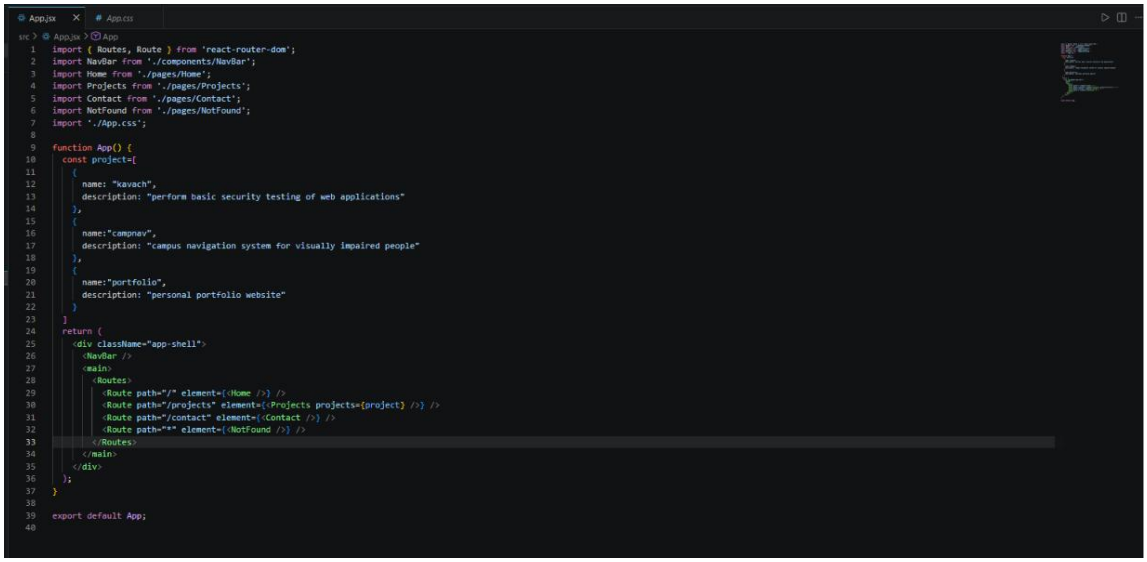
CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
FACULTY OF TECHNOLOGYAND ENGINEERING
B. TECH (IT/CE/CSE/AIML)

Subject: Advanced Web Development Frameworks (ITUE301) | Semester: 5th

Practical Statement:

This practical extends the portfolio application developed in Practical 1 by integrating React Router to enable client-side navigation. The application implements a navigation bar with at least three routes — Home, Projects, and Contact — such that each route renders a distinct component without triggering a full page reload. To demonstrate meaningful use of state management, the application incorporates at least two useState variables: one used to toggle the visibility of a UI element, and another used to manage a form input on the Contact page. Furthermore, the Contact page includes a controlled input field that captures user input and displays it in real time, ensuring the input's value stays synchronized with the component's state.

Screenshots:



```jsx
import { Routes, Route } from 'react-router-dom';
import NavBar from './components/NavBar';
import Home from './pages/Home';
import Projects from './pages/Projects';
import Contact from './pages/Contact';
import NotFound from './pages/NotFound';
import './App.css';

function App() {
  const project=[
    {
      name: "kavach",
      description: "perform basic security testing of web applications"
    },
    {
      name:"campnav",
      description: "campus navigation system for visually impaired people"
    },
    {
      name:"portfolio",
      description: "personal portfolio website"
    }
  ]
  return (
    <div className="app-shell">
      <NavBar />
      <main>
        <Routes>
          <Route path="/" element={<Home />} />
          <Route path="/projects" element={<Projects projects={project} />} />
          <Route path="/contact" element={<Contact />} />
          <Route path="*" element={<NotFound />} />
        </Routes>
      </main>
    </div>
  );
}

export default App;
```

*Figure 1The root component — defines a projects array (**kavach**: security testing for web apps; **campnav**: campus navigation for visually impaired people; **portfolio**: personal portfolio website) and sets up routing (/, /projects, /contact, and a catch-all 404 route) using react-router-dom.*

Student ID: 24IT045

App.jsx NavBar.jsx

src > components > NavBar.jsx > ...

```jsx
import { Link } from 'react-router-dom';

function NavBar() {
  return (
    <nav className="navbar">
      <ul>
        <li>
          <Link to="/">Home</Link>
        </li>
        <li>
          <Link to="/projects">Projects</Link>
        </li>
        <li>
          <Link to="/contact">Contact</Link>
        </li>
      </ul>
    </nav>
  );
}

export default NavBar;
```

Figure 1.2A navigation bar component with links (using react-router-dom) to Home, Projects, and Contact.

App.jsx Contact.jsx

src > pages > Contact.jsx > Contact

```jsx
import { useState, useEffect } from 'react';

const STORAGE_KEY = 'contactMessage';

function Contact() {
  const [message, setMessage] = useState('');
  const [savedMessage, setSavedMessage] = useState('');
  const [showHelp, setShowHelp] = useState(false);
  const [status, setStatus] = useState('');

  useEffect(() => {
    const stored = localStorage.getItem(STORAGE_KEY);
    if (stored) {
      setMessage(stored);
      setSavedMessage(stored);
    }
  }, []);

  const handleSend = () => {
    localStorage.setItem(STORAGE_KEY, message);
    setSavedMessage(message);
    setStatus('Message saved successfully.');

    window.setTimeout(() => {
      setStatus('');
    }, 2500);
  };
```

Figure 1.3 ***Contact.jsx*** *Sets up state for a message form (message, savedMessage, showHelp, status) and uses useEffect to load/save a message to localStorage..*

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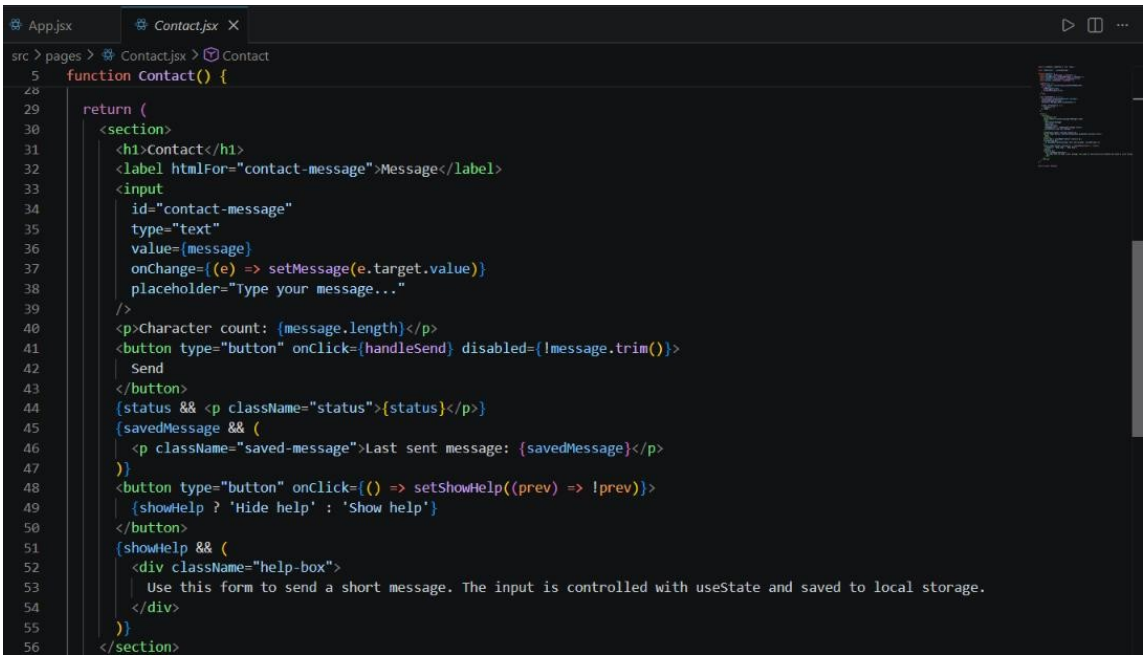


*Figure 1.4**Contact.jsx (part 2: JSX/render)***
Renders the Contact form: an input bound to message, a live character count, a Send button that saves to local storage, a "Show/Hide help" toggle, and a help box.



*Figure1.5**Home.jsx** The Home page component — welcomes the user to the "practical 2 portfolio app" and tells them to navigate using the menu above.*

Student ID: 24IT045

App.jsx NotFound.jsx

src > pages > NotFound.jsx > ...

```jsx
import { Link } from 'react-router-dom';

function NotFound() {
  return (
    <section>
      <h1>404 - Not Found</h1>
      <p>The page you are looking for does not exist.</p>
      <Link to="/">Return to home</Link>
    </section>
  );
}

export default NotFound;
```

*Figure 6**NotFound.jsx** A 404 error page with a link back to Home.*

App.jsx Projects.jsx

src > pages > Projects.jsx > Projects

```jsx
function Projects({ projects }) {
  return (
    <section>
      <h1>Projects</h1>
      <p>This page highlights sample projects and reusable components.</p>
      <ul>
        {projects.map((project, index) => (
          <li key={index}>
            <strong>{project.name}</strong>: {project.description}
          </li>
        ))}
      </ul>
    </section>
  );
}

export default Projects;
```

```
function Projects({ projects }: {
    projects: any;
}): JSX.Element
```

*Figure 1.7 **Projects.jsx** Renders a list of projects passed in as props, mapping over them to display each project's name and description.*

PROBLEMS OUTPUT DEBUG CONSOLE GITHUB PORTS TERMINAL

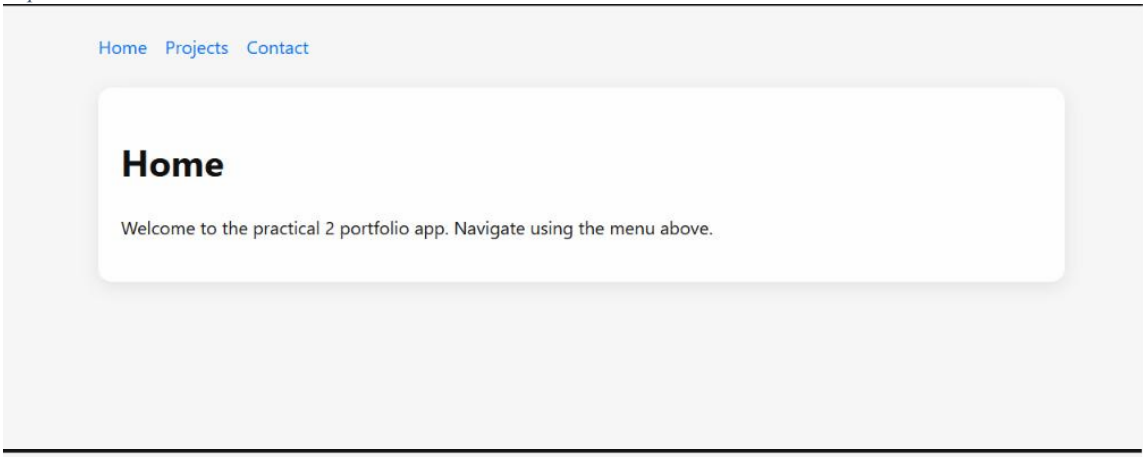
```
PS C:\sem-5\ADVANCED WEB DEVELOPMENT FRAMEWORKS\practical\p-2> npm run dev

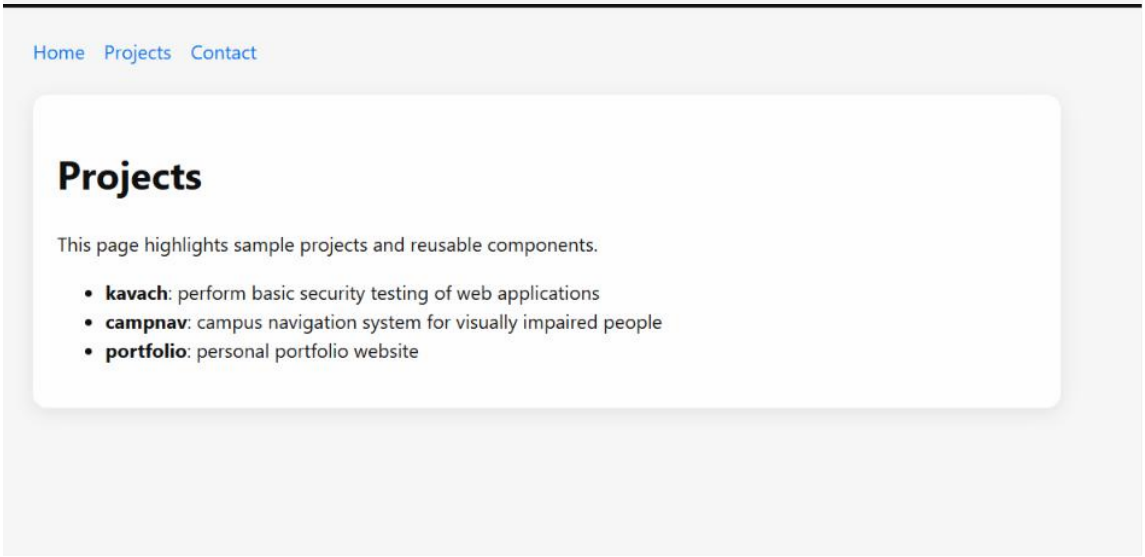
  VITE v5.4.21  ready in 429 ms

  ➜  Local:   http://localhost:5173/
  ➜  Network: use --host to expose
  ➜  press h + enter to show help
```

Student ID: 24IT045

*Figure 1.8VS Code terminal: Terminal output of npm run dev for the **p-2** project (path: sem-5\ADVANCED WEB DEVELOPMENT FRAMEWORKS\practical\p-2), started with **Vite v5.4.21**, ready in 429 ms, and served locally at http://localhost:5173/.*





Student ID: 24IT045

